



A PEACH OF A BOARD FOR DEVELOPERS

A microprocessor development board containing a printed circuit board, a microprocessor and some additional support logic serves users of the microprocessor as a method to prototype applications in products. However the thing here is that unlike regular electronic devices, a development board often contains little or no hardware dedicated to a user interface. It usually runs a dedicated program that is coded into it, and more often used by beginners. So why is the world mad about them?

These were a lucrative option for hobbyists, as you had a board set up with peripherals and could easily work as required with a few lines of code. The better part is that you could get those lines from the internet. Another feature seen in these development boards is the (read only memory) ROM-based built-in debugger.

These evolved as requirement grew and have been made really easy to use and have added some features while removing others. Beginning from manually inputting hexadecimal codes, we now have option to program these via a computer. We have a lot of development boards in the market today and it might get difficult to distinguish them at times.

Some available boards

The introduction of the arduino in 2005 made designing personal systems simpler for hobbyists and budding engineers. It consists of open source computer hardware and software. Intended for specialists, designers, and anybody intrigued by making intelligent articles, the simplicity of working with the board has made it into a means of learning with students.

The Raspberry Pi made waves due to the number of available features in the low price bracket. It can be easily plugged into a monitor, computer or your television. Also it uses standard mouse and keyboard.

BeagleBoard is another low-power, open-source hardware, single-board computer developed by a small team of engineers as an educational board for students. With the new Wi-Fi and Bluetooth additions the scope of the board has significantly increased. This includes features for Internet of Things (IoT) development.

A recent launch however has been the GR-Peach by Renesas Electronics. Let's take a closer look at the new board.

Peach processes your commands

Unlike the orangish-red tint of a regular peach, the Renesas GR-Peach RZ/

WHERE DOES IT STAND?

With all such features, is this the best there can be? Let's look at a comparative analysis among some public favourites.

Board Name	Dimensions	Processor	Power	RAM	Storage	Peripherals
GR-Peach	67.58 x 53.34 mm	ARM cortex-A9 @400 MHz	3.3V	10 MB	MicroSD	Ethernet, USB2.0, ADC, UART, GPIO, I2C, SPI
Raspberry Pi 3	85.60 x 56.5 mm	Quad core ARM cortex-A53 @ 1.2 GHz	4V	1GB	MicroSD	Ethernet, GPIO, USB 2.0, Bluetooth 4.1
Arduino 101	68.6 x 53.4 mm	Intel curie two tiny cores, an x86 (Quark) and an ARC	5V	24kB		Ethernet, USB connector, BLE, PWM o/p
Beagleboard-x15	107 x 102 mm	ARM cortex-A8@720 MHz	5V	512 MB	MicroSD	Ethernet, USB OTG, USB 2.0, HDMI, Bluetooth